

5.2 BLIP and Captioning

Filter noisy pairs, then match and decode. Bootstrap the captions.

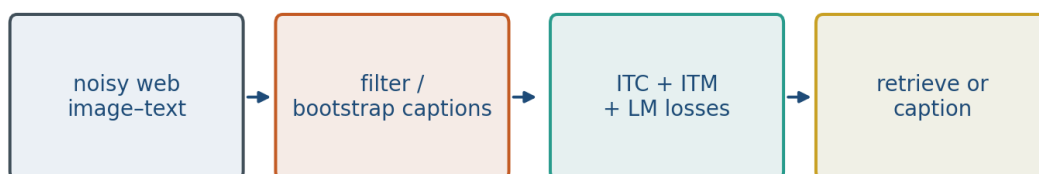
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

BLIP (Li et al., 2022) is a vision–language model that **matches** and **writes**. CLIP scores pairs. BLIP also decodes a caption, and it tries to **clean** the noisy web pairs CLIP trained on.

1. Bootstrap the data

Web alt-text is messy. BLIP’s captioner generates synthetic captions; a filter keeps the ones that still match the image (ITM: image–text matching). The cleaned pairs retrain the model. That loop is the “bootstrapped pretraining” in the syllabus.

BLIP: clean the pairs, then learn to match and to write



2. Three losses, one backbone

Loss	Job
ITC (contrastive)	CLIP-style alignment
ITM (matching)	Binary: does this caption belong to this image?
LM (language modeling)	Decode the caption, causal on text, cross-attend to the image

The image tower is a ViT. The text side is a transformer that can encode or decode depending on the head. **BLIP-2** later freezes a strong image encoder and trains a thin **Q-Former**; you can treat that as “same idea, cheaper.”

3. Captioning versus retrieval

Retrieval (CLIP) returns an existing string from a gallery. Captioning (BLIP) **generates** a new string.

Evaluation: CIDEr / CLIP-score / human, not just recall@k. Hallucinated objects are the usual failure ("a red bike" when there is none).

4. Practice

1. Why filter generated captions instead of trusting every alt-text on the web?
2. ITM vs ITC: one is a binary matcher, one is a softmax over the batch. When would ITM catch a pair ITC misses?
3. Name one metric you would report in a captioning project that CLIP retrieval would not need.